

Closed-form envelopes of the Fourier-optics kernel in aberration-corrected LEEM

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Abstract

We present closed-form spatial and chromatic envelopes of the Fourier-optics (FO) kernel for image formation in aberration-corrected low-energy electron microscopy (ac-LEEM), following Yu, Lau and Altman [11]. That work reduced the partially coherent ensemble integrals to Gaussian envelopes and compared FO with the contrast-transfer function (CTF); the closed forms were stated, not derived. After the first-order tilt, frozen-aperture and mixed tilt–energy truncations of their Eqs. (3)–(4), the remaining Gaussian integrals are identities: they yield the spatial envelope, the chromatic envelopes including the polar form of the chromatic prefactor, the FO/CTF ratios, and the Jacobi–Anger expansion of a sinusoidal phase object. The formulae are machine-checked in Lean 4 against that kernel and are the forward map inverted in the companion reconstruction [16]. This work places FO/CTF disagreement on identities of the same kernel used to predict a focal series and to recover the object that produced it.

1 Introduction

Image interpretation in LEEM [1, 2, 3, 4] requires a quantitative account of how the cathode lens, contrast aperture, source size and energy spread map the object exit wave to the recorded intensity. Aberration-corrected instruments [5, 6] make that map sharper, but they do not remove the distinction between two standard calculations. The contrast-transfer function (CTF) evaluates the inverse transforms of ΨH and $\Psi^* H^*$ separately and is computationally efficient [8, 9]. Fourier optics (FO) retains inseparable mixing of spatial-frequency pairs $(\mathbf{q}, \mathbf{q}')$ under partial coherence [7, 10, 11]. The mathematically rigorous FO calculation gains an advantage under conditions, particularly defocus, in which partial coherence compromises the validity of the CTF intensity [11]. That distinction is observed for strongly rippled graphene, where CTF modelling fails and FO reproduces the measured through-focal contrast [10].

Yu, Lau and Altman [11] write the ensemble-averaged kernel as an integral of a coherent kernel against a Gaussian source density and a Gaussian energy density, extending Pang *et al.* [7] to aberration-corrected columns (C_5 , C_{CC} , C_{3C} in place of C_3 , C_C). Closed-form envelopes—their Eq. (5), Eqs. (6a)–(6b), the polar form of the chromatic prefactor, and the FO/CTF ratios Γ_S and Γ_C —are stated in an appendix. They enter every FO intensity used to interpret a focal series, and they are the forward kernel of the companion through-focal inversion [16]. They were not derived. Here those integrals are evaluated. After the truncations of paper Eqs. (3)–(4) they are identities; the truncations themselves are modelling hypotheses. Selected identities are machine-checked in Lean 4 [14, 15] against that kernel (Appendix A). Experimental claims and the numerical accuracy of Table 1 in [11] are not claimed.

2 Image formation

The object is the complex exit wave $\psi_0(\mathbf{r}) = \sigma(\mathbf{r}) e^{i\phi(\mathbf{r})}$, with ϕ in radians. Spatial frequency is $\mathbf{q} = \boldsymbol{\alpha}/\lambda$, where $\boldsymbol{\alpha}$ is the emission-angle vector from the virtual object and λ is the wavelength

after acceleration to column energy E [9, 8]. Vectors are bold and $q = |\mathbf{q}|$; in one dimension q, q', k are signed scalars, so qq' is a product of signed reals. The Fourier kernel is the unitary 2π -in-the-exponent convention $e^{2\pi i \mathbf{q} \cdot \mathbf{r}}$, with $d^2 q = dq_x dq_y$. Wave aberration χ is measured in waves, with transfer factor $W = e^{2\pi i \chi}$. Object phase is independent of that 2π convention. The principal branch of the complex logarithm has argument in $(-\pi, \pi]$, and $z^{1/2} = \exp(\frac{1}{2} \text{Log } z)$ with $z^{1/2} > 0$ for $z > 0$ real.

With $\Psi = \mathcal{F}[\psi_0]$, the FO intensity is bilinear in the object spectrum [12, 7, 11],

$$I(\mathbf{r}, \Delta z) = \iint \Psi(\mathbf{q}) \Psi^*(\mathbf{q}') R(\mathbf{q}, \mathbf{q}', \Delta z) e^{2\pi i (\mathbf{q} - \mathbf{q}') \cdot \mathbf{r}} d^2 q d^2 q'. \quad (1)$$

A circular contrast aperture of cutoff $q_{\text{ap}} = \alpha_{\text{ap}}/\lambda$ is the indicator $M(\mathbf{q}) = \mathbf{1}_{\{|\mathbf{q}| \leq q_{\text{ap}}\}}$, which is real.

Approximation 2.1 (Frozen aperture). For a small source, $M(\mathbf{q} + \mathbf{k}) \approx M(\mathbf{q})$. This is a modelling hypothesis, not an identity.

Spherical aberration and defocus enter as

$$\chi_S(\mathbf{q}, \Delta z) = \frac{1}{4} C_3 \lambda^3 q^4 + \frac{1}{6} C_5 \lambda^5 q^6 + \frac{1}{2} \Delta z \lambda q^2, \quad (2)$$

with $W_S = e^{2\pi i \chi_S}$. The sign of Δz is the Tromp–Schramm convention: $\Delta z > 0$ adds positive optical path for off-axis rays [9]. Mixing the opposite sign used in some simulators interchanges overfocus and underfocus. Non-corrected (nac) LEEM keeps C_3 and drops C_5 ; aberration-corrected (ac) LEEM sets $C_3 = 0$ and keeps C_5 . Chromatic aberration is

$$\chi_C(\mathbf{q}, \varepsilon) = \frac{1}{2} C_C \lambda \left(\frac{\varepsilon}{E}\right) q^2 + \frac{1}{2} C_{CC} \lambda \left(\frac{\varepsilon}{E}\right)^2 q^2 + \frac{1}{4} C_{3C} \lambda^3 \left(\frac{\varepsilon}{E}\right) q^4, \quad (3)$$

with $W_C = e^{2\pi i \chi_C}$. Nac keeps C_C and drops C_{CC}, C_{3C} ; ac sets $C_C = 0$ and keeps C_{CC}, C_{3C} . If χ is real then $W(\mathbf{q})W^*(\mathbf{q}') = e^{2\pi i (\chi(\mathbf{q}) - \chi(\mathbf{q}'))}$; at $\varepsilon = 0$, $\chi_C(\cdot, 0) = 0$ and $W_C(\cdot, 0) = 1$.

Viewing (2) as a radial function on $\mathbb{R}^2$,

$$\nabla_{\mathbf{q}} \chi_S(\mathbf{q}, \Delta z) = (C_3 \lambda^3 q^2 + C_5 \lambda^5 q^4 + \Delta z \lambda) \mathbf{q}. \quad (4)$$

In one dimension, $\partial_q \chi_S = C_3 \lambda^3 q^3 + C_5 \lambda^5 q^5 + \Delta z \lambda q$, so odd powers keep the sign of q .

The coherent kernel is paper Eq. (2) [11],

$$R_0(\mathbf{q}, \mathbf{q}', \Delta z, \varepsilon) = M(\mathbf{q}) M(\mathbf{q}') W_S(\mathbf{q}, \Delta z) W_S^*(\mathbf{q}', \Delta z) W_C(\mathbf{q}, \varepsilon) W_C^*(\mathbf{q}', \varepsilon), \quad (5)$$

and the ensemble-averaged kernel is paper Eq. (1),

$$R(\mathbf{q}, \mathbf{q}', \Delta z) = \int_{\mathbb{R}^2} d^2 k \int_{\mathbb{R}} d\varepsilon s(\mathbf{k}) c(\varepsilon) R_0(\mathbf{q} + \mathbf{k}, \mathbf{q}' + \mathbf{k}, \Delta z, \varepsilon). \quad (6)$$

The source and energy densities are normalised Gaussians of widths $\sigma_{\text{ill}} > 0$ and $\sigma_E > 0$,

$$\begin{aligned} c(\varepsilon) &= \frac{1}{\sigma_E \sqrt{2\pi}} \exp\left(-\frac{\varepsilon^2}{2\sigma_E^2}\right), \\ s_{1D}(k) &= \frac{1}{\sigma_{\text{ill}} \sqrt{2\pi}} \exp\left(-\frac{k^2}{2\sigma_{\text{ill}}^2}\right), \quad s(\mathbf{k}) = s_{1D}(k_x) s_{1D}(k_y). \end{aligned} \quad (7)$$

The paper identifies ΔE with the FWHM of c and $q_{\text{ill}} = \alpha_{\text{ill}}/\lambda$ with the isotropic FWHM of s . A centred Gaussian $f(x) = \exp(-x^2/(2\sigma^2))$ has FWHM $\Delta = 2\sqrt{2 \ln 2} \sigma$, hence $\sigma^2 = \Delta^2/(8 \ln 2)$ (Lemma A.2).

Define the tilt vector

$$\mathbf{a}(\mathbf{q}, \mathbf{q}', \Delta z) := \nabla \chi_S(\mathbf{q}, \Delta z) - \nabla \chi_S(\mathbf{q}', \Delta z). \quad (8)$$

On the diagonal $\mathbf{a}(\mathbf{q}, \mathbf{q}) = 0$; on the axis $\mathbf{q}' = \mathbf{0}$, $\mathbf{a}(\mathbf{q}, \mathbf{0}) = \nabla \chi_S(\mathbf{q})$. If $E \neq 0$, the chromatic path difference splits exactly as $\chi_C(\mathbf{q}, \varepsilon) - \chi_C(\mathbf{q}', \varepsilon) = b_1 \varepsilon + b_2 \varepsilon^2$, with

$$\begin{aligned} b_1(\mathbf{q}, \mathbf{q}') &= \frac{C_C \lambda}{2E} (q^2 - q'^2) + \frac{C_{3C} \lambda^3}{4E} (q^4 - q'^4), \\ b_2(\mathbf{q}, \mathbf{q}') &= \frac{C_{CC} \lambda}{2E^2} (q^2 - q'^2). \end{aligned} \quad (9)$$

This is an identity: no other powers of ε appear. On the diagonal $b_1 = b_2 = 0$; both coefficients are antisymmetric in $(\mathbf{q}, \mathbf{q}')$. Under ac, $C_C = 0$ and $b_1 = (C_{3C} \lambda^3 / (4E)) (q^4 - q'^4)$; under nac, $C_{CC} = C_{3C} = 0$, so $b_2 = 0$.

Approximation 2.2 (First-order tilt, no mixed $k\varepsilon$). Expand the path functions to first order in $\mathbf{k}$ and drop mixed tilt–energy terms (paper Eq. (3)):

$$\chi_S(\mathbf{q} + \mathbf{k}, \Delta z) + \chi_C(\mathbf{q} + \mathbf{k}, \varepsilon) \approx \chi_S(\mathbf{q}, \Delta z) + \mathbf{k} \cdot \nabla \chi_S(\mathbf{q}, \Delta z) + \chi_C(\mathbf{q}, \varepsilon). \quad (10)$$

Together with [Approximation 2.1](#) this is paper Eq. (4) [11]:

$$R_0(\mathbf{q} + \mathbf{k}, \mathbf{q}' + \mathbf{k}, \Delta z, \varepsilon) \approx R_0(\mathbf{q}, \mathbf{q}', \Delta z, 0) e^{2\pi i \mathbf{a} \cdot \mathbf{k}} e^{2\pi i (b_1 \varepsilon + b_2 \varepsilon^2)}. \quad (11)$$

The $(\mathbf{k}, \varepsilon)$ integral then factors as $E_S E_{C,\text{tot}}$.

Because χ_S is a polynomial the Taylor remainder is $O(|\mathbf{k}|^2)$; (10) does not justify dropping that remainder uniformly in $\mathbf{q}$.

Under [Approximation 2.2](#) the working kernel is

$$R(\mathbf{q}, \mathbf{q}', \Delta z) = R_0(\mathbf{q}, \mathbf{q}', \Delta z, 0) E_S(\mathbf{q}, \mathbf{q}', \Delta z) E_{C,\text{tot}}(\mathbf{q}, \mathbf{q}'), \quad (12)$$

with E_S from (16) and $E_{C,\text{tot}}$ from (20). Nac/ac specialisation is substitution of coefficients, not a new identity.

3 Spatial envelope

After [Approximation 2.2](#) the spatial envelope is

$$E_S(\mathbf{q}, \mathbf{q}', \Delta z) := \int_{\mathbb{R}^2} s(\mathbf{k}) e^{2\pi i \mathbf{a} \cdot \mathbf{k}} d^2 k. \quad (13)$$

The characteristic function of a centred Gaussian of width σ is

$$\int_{\mathbb{R}} g_\sigma(k) e^{2\pi i a k} dk = \exp(-2\pi^2 \sigma^2 a^2), \quad (14)$$

and the two-dimensional source factors, so

$$\int_{\mathbb{R}^2} s(\mathbf{k}) e^{2\pi i \mathbf{a} \cdot \mathbf{k}} d^2 k = \exp(-2\pi^2 \sigma_{\text{ill}}^2 |\mathbf{a}|^2) \quad (15)$$

([Lemma A.1](#) and [Theorem A.3](#)). If $q_{\text{ill}} > 0$ and $\kappa = \pi^2 q_{\text{ill}}^2 / (4 \ln 2)$, the FWHM conversion $\sigma_{\text{ill}}^2 = q_{\text{ill}}^2 / (8 \ln 2)$ rewrites (15) as paper Eq. (5),

$$E_S(\mathbf{q}, \mathbf{q}', \Delta z) = \exp\left(-\frac{\pi^2 q_{\text{ill}}^2}{4 \ln 2} |\mathbf{a}(\mathbf{q}, \mathbf{q}', \Delta z)|^2\right) = e^{-\kappa |\mathbf{a}|^2}. \quad (16)$$

The envelope is real, even under $(\mathbf{q}, \mathbf{q}') \leftrightarrow (\mathbf{q}', \mathbf{q})$, and equals 1 on the diagonal. Thus $\kappa = 2\pi^2\sigma_{\text{ill}}^2$, not $\pi^2\sigma^2$ (a factor of two); substituting FWHM for σ instead of $\sigma^2 = \Delta^2/(8 \ln 2)$ is a factor $8 \ln 2$.

Writing $u(\mathbf{q}) := C_3\lambda^3 q^2 + C_5\lambda^5 q^4 + \Delta z\lambda$, one has $\nabla\chi_S = u(\mathbf{q})\mathbf{q}$ and $\mathbf{a} = u(\mathbf{q})\mathbf{q} - u(\mathbf{q}')\mathbf{q}'$, hence $|\mathbf{a}|^2 = u(\mathbf{q})^2 q^2 + u(\mathbf{q}')^2 q'^2 - 2u(\mathbf{q})u(\mathbf{q}')(\mathbf{q} \cdot \mathbf{q}')$. The inner product $\mathbf{q} \cdot \mathbf{q}'$ cannot be replaced by $q q'$. In one dimension (16) is

$$E_S(q, q', \Delta z) = \exp\left(-\kappa\left[C_3\lambda^3(q^3 - q'^3) + C_5\lambda^5(q^5 - q'^5) + \Delta z\lambda(q - q')\right]^2\right); \quad (17)$$

one may factor $(q - q')^2$ via $q^3 - q'^3 = (q - q')(q^2 + qq' + q'^2)$. For ac, $C_3 = 0$. The CTF slice is $E_S(\mathbf{q}, \mathbf{0}, \Delta z) = \exp(-\kappa|\nabla\chi_S(\mathbf{q})|^2)$.

4 Chromatic envelope

The chromatic envelope is

$$E_{C,\text{tot}}(\mathbf{q}, \mathbf{q}') := \int_{\mathbb{R}} c(\varepsilon) \exp(2\pi i(b_1\varepsilon + b_2\varepsilon^2)) d\varepsilon. \quad (18)$$

The integrand is a constant times $\exp(B\varepsilon^2 + C\varepsilon)$ with $B = -1/(2\sigma_E^2) + 2\pi i b_2$ and $C = 2\pi i b_1$. If $\sigma_E > 0$ then $\text{Re}(B) = -1/(2\sigma_E^2) < 0$, independently of b_2 : Gaussian decay comes only from the real energy density. Writing $z := 1 - 4\pi i b_2 \sigma_E^2$ one has $\text{Re}(z) = 1 > 0$ and $B = -z/(2\sigma_E^2)$. Evaluating Lemma A.1 on the principal branch (Theorem A.5) produces paper Eqs. (6a)–(6b),

$$E_{CC}(\mathbf{q}, \mathbf{q}') := (1 - 4\pi i b_2 \sigma_E^2)^{-1/2} = z^{-1/2}, \quad (19)$$

with value 1 at $b_2 = 0$, and

$$E_{C,\text{tot}} = E_{CC} \exp(-2\pi^2 \sigma_E^2 b_1^2 E_{CC}^2) = (1 - 4\pi i b_2 \sigma_E^2)^{-1/2} \exp\left(-\frac{2\pi^2 \sigma_E^2 b_1^2}{1 - 4\pi i b_2 \sigma_E^2}\right). \quad (20)$$

In FWHM variables, $z = 1 - i(\pi b_2 \Delta E^2)/(2 \ln 2)$. The Hanßen–Trepte packaging [13] writes $2\pi^2 \sigma_E^2 b_1^2 = (\Delta E)^2 (2\pi b_1)^2 / (16 \ln 2)$, where $2\pi b_1$ is the coefficient of ε in the radian phase $2\pi \Delta \chi_C$: both π (from $W = e^{2\pi i \chi}$) and $16 \ln 2$ (from FWHM) must appear.

If $b_2 = 0$ then $E_{CC} = 1$ and $E_{C,\text{tot}} = \exp(-2\pi^2 \sigma_E^2 b_1^2)$. On the nac CTF slice $\mathbf{q}' = \mathbf{0}$, $b_1 = C_C \lambda q^2 / (2E)$, hence

$$E_{C,\text{tot}}(\mathbf{q}, \mathbf{0}) = \exp\left(-\frac{(\pi C_C \lambda \Delta E q^2)^2}{16 \ln 2 E^2}\right), \quad (21)$$

the standard chromatic envelope of Schramm *et al.* [8]. The density c is real, so $E_{C,\text{tot}}(\mathbf{q}', \mathbf{q}) = \overline{E_{C,\text{tot}}(\mathbf{q}, \mathbf{q}')}$.

4.1 Polar form

Write $y := 4\pi b_2 \sigma_E^2 = \pi b_2 \Delta E^2 / (2 \ln 2)$, so $E_{CC} = (1 - iy)^{-1/2}$. For $y \in \mathbb{R}$, $|1 - iy| = \sqrt{1 + y^2}$ and $\text{Arg}(1 - iy) = -\arctan y$: the real part is identically 1, so the argument never jumps. On the principal branch,

$$(1 - iy)^{-1/2} = (1 + y^2)^{-1/4} \exp\left(\frac{i}{2} \arctan y\right). \quad (22)$$

The phase of E_{CC} therefore lies in $(-\pi/4, \pi/4)$. Under ac, $y = \gamma(q^2 - q'^2)$ with

$$\gamma := \frac{\pi C_{CC} \lambda \Delta E^2}{4 \ln 2 E^2}, \quad (23)$$

and (22) is $E_{CC} = [1 + \gamma^2(q^2 - q'^2)^2]^{-1/4} \exp(\frac{i}{2} \arctan(\gamma(q^2 - q'^2)))$.

Let

$$\eta := \frac{(\pi C_3 \lambda^3 \Delta E)^2}{64 \ln 2 E^2}, \quad (24)$$

so that $2\pi^2 \sigma_E^2 b_1^2 = \eta(q^4 - q'^4)^2$ under ac. Rationalising $1/(1 - iy) = (1 + iy)/(1 + y^2)$ splits the remaining exponential:

$$\begin{aligned} \exp(-2\pi^2 \sigma_E^2 b_1^2 E_{CC}^2) &= \exp\left(-\frac{\eta(q^4 - q'^4)^2}{1 + \gamma^2(q^2 - q'^2)^2}\right) \\ &\cdot \exp\left(-i \frac{\eta \gamma (q^2 - q'^2)(q^4 - q'^4)^2}{1 + \gamma^2(q^2 - q'^2)^2}\right). \end{aligned} \quad (25)$$

Writing $E_{C,\text{tot}} = Ae^{i\theta}$ with $A \geq 0$, moduli multiply and arguments add:

$$\begin{aligned} A(q, q') &= [1 + \gamma^2(q^2 - q'^2)^2]^{-1/4} \exp\left(-\frac{\eta(q^4 - q'^4)^2}{1 + \gamma^2(q^2 - q'^2)^2}\right), \\ \theta(q, q') &= \frac{1}{2} \arctan(\gamma(q^2 - q'^2)) - \frac{\eta \gamma (q^2 - q'^2)(q^4 - q'^4)^2}{1 + \gamma^2(q^2 - q'^2)^2}. \end{aligned} \quad (26)$$

Parity: $A(q, q') = A(q', q) = A(|q|, |q'|)$ and $\theta(q, q') = -\theta(q', q)$.

5 Comparison with the contrast-transfer function

The paper compares FO with a separable CTF kernel through the ratios [11]

$$\begin{aligned} \Gamma_C(\mathbf{q}, \mathbf{q}') &= \frac{E_{C,\text{tot}}(\mathbf{q}, \mathbf{0}) E_{C,\text{tot}}(\mathbf{q}', \mathbf{0})}{E_{C,\text{tot}}(\mathbf{q}, \mathbf{q}')} , \\ \Gamma_S(\mathbf{q}, \mathbf{q}', \Delta z) &= \frac{E_S(\mathbf{q}, \mathbf{0}, \Delta z) E_S(\mathbf{q}', \mathbf{0}, \Delta z)}{E_S(\mathbf{q}, \mathbf{q}', \Delta z)}. \end{aligned} \quad (27)$$

Printed (7a) uses the non-conjugated product $E_{C,\text{tot}}(\mathbf{q}, \mathbf{0}) E_{C,\text{tot}}(\mathbf{q}', \mathbf{0})$. For ac, $E_{C,\text{tot}}$ is complex, so Γ_C is complex; the plotted quantity is $|\Gamma_C|$. In the polar form of Section 4.1,

$$\Gamma_C(q, q') = \frac{A(q, 0) A(q', 0)}{A(q, q')} \exp(i[\theta(q, 0) + \theta(q', 0) - \theta(q, q')]). \quad (28)$$

Let $\mathbf{u} = \nabla \chi_S(\mathbf{q}, \Delta z)$ and $\mathbf{v} = \nabla \chi_S(\mathbf{q}', \Delta z)$. Then $E_S = e^{-\kappa|\mathbf{u}-\mathbf{v}|^2}$ and the quadratic terms cancel, leaving

$$\Gamma_S = \exp(-2\kappa \mathbf{u} \cdot \mathbf{v}). \quad (29)$$

The inner product is $\nabla \chi_S(\mathbf{q}) \cdot \nabla \chi_S(\mathbf{q}')$, not a product of Euclidean norms. In one dimension with $C_3 = 0$,

$$\Gamma_S(q, q', \Delta z) = \exp\left[-\frac{\pi^2 q_{\text{ill}}^2}{2 \ln 2} (C_5 \lambda^5 q^5 + \Delta z \lambda q) (C_5 \lambda^5 q'^5 + \Delta z \lambda q')\right]. \quad (30)$$

If $q_{\text{ill}} = \Delta E = 0$ then every envelope equals 1, hence $\Gamma_C = \Gamma_S = 1$. On either spatial axis $\Gamma_S = 1$. The chromatic ratio satisfies $\Gamma_C(\mathbf{q}, \mathbf{0}) = 1$; if $\Delta E > 0$ then only $|\Gamma_C(\mathbf{0}, \mathbf{q}')| = 1$ in general for ac, because (27) has no extra conjugate.

After Approximation 2.2 the working kernels are

$$R_{\text{FO}}(\mathbf{q}, \mathbf{q}', \Delta z) := R_0(\mathbf{q}, \mathbf{q}', \Delta z, 0) E_S(\mathbf{q}, \mathbf{q}', \Delta z) E_{C,\text{tot}}(\mathbf{q}, \mathbf{q}'), \quad (31)$$

$$R_{\text{CTF}}(\mathbf{q}, \mathbf{q}', \Delta z) := R_0(\mathbf{q}, \mathbf{q}', \Delta z, 0) E_S(\mathbf{q}, \mathbf{0}, \Delta z) E_S(\mathbf{q}', \mathbf{0}, \Delta z) E_{C,\text{tot}}(\mathbf{q}, \mathbf{0}) E_{C,\text{tot}}(\mathbf{q}', \mathbf{0}), \quad (32)$$

where the last two chromatic factors are not conjugated, matching printed (7a). Neither envelope vanishes (E_S is an exponential; $E_{CC} = z^{-1/2}$ with $\text{Re}(z) = 1$), so multiplying (31) by (27) yields the comparison identity

$$R_{\text{CTF}} = R_{\text{FO}} \Gamma_C \Gamma_S. \quad (33)$$

On $\mathbf{q}' = \mathbf{0}$ one has $\Gamma_C = \Gamma_S = 1$, hence $R_{\text{FO}}(\mathbf{q}, \mathbf{0}, \Delta z) = R_{\text{CTF}}(\mathbf{q}, \mathbf{0}, \Delta z)$. Thus $\Gamma_C \Gamma_S \neq 1$ is the warning that a separable CTF kernel differs from the FO kernel; the two agree on the axes, at perfect coherence, and in the weak-phase limit that samples those axes. The coherent kernel is Hermitian, $\overline{R_0(\mathbf{q}, \mathbf{q}', \Delta z, \varepsilon)} = R_0(\mathbf{q}', \mathbf{q}, \Delta z, \varepsilon)$, because M is real and each of W_S and W_C is unimodular.

6 Sinusoidal phase object

A kinematic LEEM phase object of physical amplitude A and wavelength Λ is $\psi_0(x) \propto \exp(i\varphi \sin(kx))$ with $\varphi = 4\pi A/\lambda_0$ and $k = 2\pi/\Lambda$ [10, 11]. The phase amplitude is a definition of the reflection kinematics, not an FO theorem. At $E_0 = 10$ eV, $\lambda_0 \simeq 0.39$ nm, so $A = 0.1$ nm already gives $\varphi \sim 3$, outside the weak-phase regime occupied by graphene ripples and atomic steps [10]. The Bessel coefficients are the Fourier coefficients

$$J_n(\varphi) = \frac{1}{2\pi} \int_{(0, 2\pi]} \exp(i\varphi \sin \theta) e^{-in\theta} d\theta, \quad (34)$$

and Jacobi–Anger supplies the expansion

$$\exp(i\varphi \sin \theta) = \sum_{n \in \mathbb{Z}} J_n(\varphi) e^{in\theta}, \quad (35)$$

absolutely and uniformly on $\mathbb{R}$ (Theorem A.6), hence $\exp(i\varphi \sin(kx)) = \sum_n J_n(\varphi) e^{inkx}$. In the paper’s 2π Fourier convention, $e^{inkx} = e^{2\pi i q_n x}$ with $q_n = n/\Lambda$. Larger φ populates larger $|n|$, so pairs (q, q') leave the axes where $\Gamma_C \Gamma_S = 1$, and a separable CTF becomes inaccurate. That is the object used in [10, 11] to exhibit CTF failure, and the aperture-truncated Bessel ladder inverted in the companion reconstruction [16].

7 Conclusions

We evaluate the closed-form spatial and chromatic envelopes of the FO kernel of Yu, Lau and Altman [11]. After the truncations of their Eqs. (3)–(4) those envelopes are identities: the spatial envelope is the characteristic function of a Gaussian source, and the chromatic envelopes follow from a complex Gaussian integral, including the polar form of the chromatic prefactor. The comparison $R_{\text{CTF}} = R_{\text{FO}} \Gamma_C \Gamma_S$ isolates the off-axis pairs at which FO and CTF disagree; the two agree on the axes, at perfect coherence, and for weak-phase objects that sample those axes. Jacobi–Anger places a sinusoidal object on the frequency ladder where that distinction matters. These identities are the forward kernel inverted in the companion through-focal reconstruction, with selected claims machine-checked in Lean 4 against that kernel. FO/CTF disagreement is therefore an identity of the same kernel used to predict a focal series and to recover the object that produced it.

A Formal claims

The statements below are the analytic identities used in the main text. Approximations 2.1 and 2.2 are not theorems: the working kernel R_{FO} is defined as the factorised expression after those truncations. Wavelength is the field `1am` (Lean reserves λ). No experimental claim or Table 1 numerical accuracy is asserted.

Lemma A.1 (Complex Gaussian integral). *Let $B, C, D \in \mathbb{C}$ with $\operatorname{Re}(B) < 0$. Then*

$$\int_{\mathbb{R}} \exp(Bx^2 + Cx + D) dx = \left(\frac{\pi}{-B}\right)^{1/2} \exp\left(D - \frac{C^2}{4B}\right), \quad (36)$$

using the principal square root.

Proof. Complete the square: $Bx^2 + Cx + D = B(x + C/(2B))^2 + D - C^2/(4B)$. The hypothesis $\operatorname{Re}(B) < 0$ gives Gaussian decay in a neighbourhood of the real axis, so the contour may be shifted. Because $\operatorname{Re}(-B) > 0$, one has $\operatorname{Arg}(-B) \in (-\pi/2, \pi/2)$ and the principal square root never meets its branch cut. When B is real and negative this recovers $\sqrt{\pi/|B|}$. $\square$

Lemma A.2 (FWHM of a centred Gaussian). *Let $f(x) = \exp(-x^2/(2\sigma^2))$ with $\sigma > 0$. If $f(\pm\Delta/2) = \frac{1}{2}f(0)$, then $\Delta = 2\sqrt{2\ln 2}\sigma$ and $\sigma^2 = \Delta^2/(8\ln 2)$.*

Proof. $f(0) = 1$, so $\exp(-(\Delta/2)^2/(2\sigma^2)) = 1/2$. Taking $-\log$ yields $\Delta^2/(8\sigma^2) = \ln 2$. $\square$

Theorem A.3 (Characteristic functions). *For $\sigma > 0$ and $a \in \mathbb{R}$, (14) holds; for $\mathbf{a} \in \mathbb{R}^2$, (15) holds with s as in (7).*

Proof. The one-dimensional integrand is a constant times $\exp(Bk^2 + Ck)$ with $B = -1/(2\sigma^2)$ and $C = 2\pi ia$. Lemma A.1 cancels the Gaussian normalisation (positive square root, $\sigma > 0$) and produces the exponent $-C^2/(4B)$. Then $C^2 = -4\pi^2 a^2$, so $-C^2/(4B) = -2\pi^2 \sigma^2 a^2$. The two-dimensional source is a product of one-dimensional Gaussians of the same width, and $\mathbf{a} \cdot \mathbf{k}$ factors the exponential, so the exponents add. $\square$

Lemma A.4 (Square-root identity). *If $A > 0$ is real and $D \in \mathbb{C} \setminus \{0\}$ has $\operatorname{Arg}(D) \neq \pi$, then $(A/D)^{1/2} = A^{1/2}D^{-1/2}$ on principal branches. This applies to $z = 1 - 4\pi ib_2\sigma_E^2$, because $\operatorname{Re}(z) = 1$.*

The identity $\operatorname{Log}(1/w) = -\operatorname{Log} w$ fails on the negative reals; $\operatorname{Re}(z) = 1$ keeps the cut away, so $b_2 = 0$ yields $+1$.

Theorem A.5 (Paper Eqs. (6a)–(6b)). *If $\sigma_E > 0$, then (19)–(20) hold on the principal branch.*

Proof. Lemma A.1 produces the prefactor $(1/(\sigma_E\sqrt{2\pi}))(\pi/(-B))^{1/2}$. Then $\pi/(-B) = 2\pi\sigma_E^2/z$. Lemma A.4 cancels the Gaussian normalisation, leaving $z^{-1/2} = E_{CC}$. At $b_2 = 0$, $z = 1$ and $1^{-1/2} = 1$ (not -1). The exponent is $-C^2/(4B)$ with $C = 2\pi ib_1$, hence $C^2 = -4\pi^2 b_1^2$ and $-C^2/(4B) = -2\pi^2 \sigma_E^2 b_1^2/z$. Forgetting $i^2 = -1$ produces a purely oscillatory “envelope”. Since $E_{CC}^2 = z^{-1}$ this is $-2\pi^2 \sigma_E^2 b_1^2 E_{CC}^2$. $\square$

Theorem A.6 (Jacobi–Anger). *For all real φ, θ , (35) converges absolutely and uniformly on $\mathbb{R}$.*

Proof. The map $\theta \mapsto \exp(i\varphi \sin \theta)$ is continuous, 2π -periodic and unimodular; its second derivative is bounded by $|\varphi| + \varphi^2$. Two integrations by parts on the circle bound $|J_n(\varphi)| \leq (|\varphi| + \varphi^2)/n^2$ for $n \neq 0$, so the coefficients are absolutely summable. A continuous function on the circle with absolutely summable Fourier coefficients equals its Fourier series pointwise; both sides of (35) are 2π -periodic, so the identity extends to all real θ . $\square$

Lean correspondence. The identities above are encoded without `sorry` in this repository, using Mathlib’s Gaussian and complex-power lemmas.

Claim	Lean
FWHM / $\sigma^2 = \Delta^2/(8 \ln 2)$	<code>gaussian_fwhm</code>
Char. fun. $E[e^{2\pi i a k}]$ (A1)	<code>charFun_source1D</code> , <code>charFun_source2D</code>
E_S integral = closed form (5)	<code>spatialEnvelopeIntegral_eq_closed</code>
$\nabla \chi_S / \partial_q \chi_S$	<code>hasDerivAt_chiS</code> , <code>hasGradientAt_chiS2</code>
Taylor remainder $o(k)$ (3)	<code>chiS_taylor</code>
$b_1 \varepsilon + b_2 \varepsilon^2$	<code>chiC_sub</code>
$E_S = e^{-\kappa a^2}$ (5)	<code>spatialEnvelope_eq_fwhm</code>
$E_{C,\text{tot}}$ closed form (6a)–(6b)	<code>chromaticEnvelopeIntegral_eq_closed</code>
Polar E_{CC} ; $\text{Arg}(1 - iy)$ (A2)	<code>ecc_eq_polar</code> , <code>arg_one_sub_I</code>
Polar A, θ of $E_{C,\text{tot}}$ (A2)	algebra of <code>ecc_eq_polar</code>
$\Gamma_S = e^{-2\kappa uv}$ (7b)	<code>gammaS_eq_kappa</code> , <code>gammaS_eq_dot</code>
Axes / perfect coherence	<code>gammaS_axis</code> , <code>gammaC_axis</code>
FO recovers CTF on $\mathbf{q}' = 0$	<code>R_F0_eq_R_CTF_axis</code>
$R_{\text{CTF}} = R_{\text{FO}} \Gamma_C \Gamma_S$ (7)	<code>R_CTF_eq_R_F0_mul_gamma</code>
Jacobi–Anger (§3.2)	<code>jacobi_anger</code>
Frozen aperture / first-order tilt (3)–(4)	not a Lean theorem
Mixed tilt–energy terms (3)–(4)	dropped by definition of R_{FO}
Table 1 numerical accuracy	informal

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